

LONDON DISCONNECTED

A spatial perspective on Internet speed, usage, and deprivation



Source: vodafone.co.uk

Introduction

In October 2024, Vodafone unveiled a provocative 3D mural in Central London, *Mind the Connectivity Gap*, cleverly referencing the famous London Underground phrase. This public artwork draws attention to the “digital divide” of some areas receiving strong Internet infrastructure, while millions lack adequate connectivity.

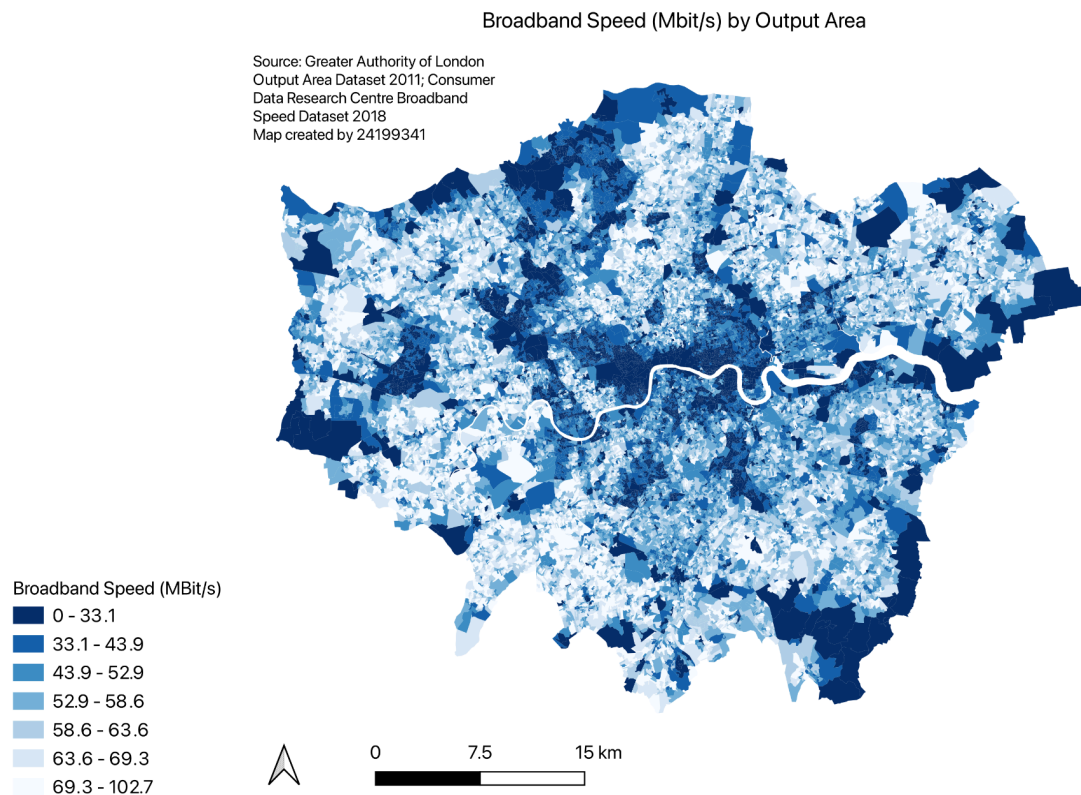
In the UK, broadband rollout is driven by private sector investment, supported by policies to incentivize companies to expand their networks. This market-driven approach prioritizes profitable urban centers and affluent neighborhoods, often leaving lower-income and rural areas with slower speeds (Holmes and Burgess, 2022). While government initiatives such as Project Gigabit subsidize connectivity in less commercially viable areas, gaps remain in regions deemed less profitable (gov.uk). Common metrics such as the Index of Multiple Deprivation (IMD), while comprehensive, may be limited when accounting for the role of digital access in modern life.

This project will analyze spatial disparities in broadband speeds and Internet usage patterns within London. By integrating these findings with the IMD, it will evaluate whether digital exclusion should become a more explicit factor in assessing deprivation, advocating for updated methodologies to reflect the realities of the digital age.

Broadband Speed and Internet User Classification

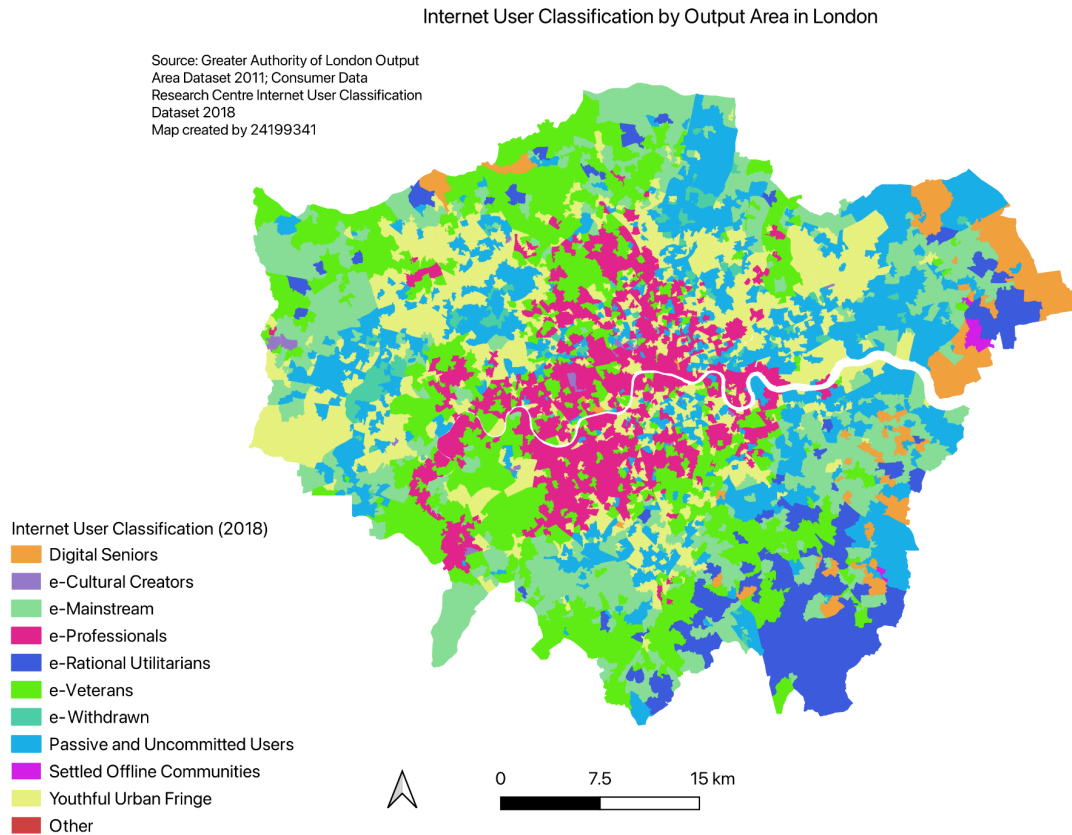
First, we will compare the average broadband speed to the Internet User Classification per output area. This will provide insight into how speeds may differ by the demand of the output area.

Figure 1: Broadband Speed Averaged by Output Area



As seen in Figure 1, there are concentrations of areas with low broadband speeds in Central London and in the Southeast and Western ends of greater London. The highest speeds are concentrated in some Northeast and Southwest areas of greater London. Aside from certain regions in the outskirts of London, a trend exists of Internet speeds getting gradually stronger away from Central London.

Figure 2: Internet User Classification by Output Area



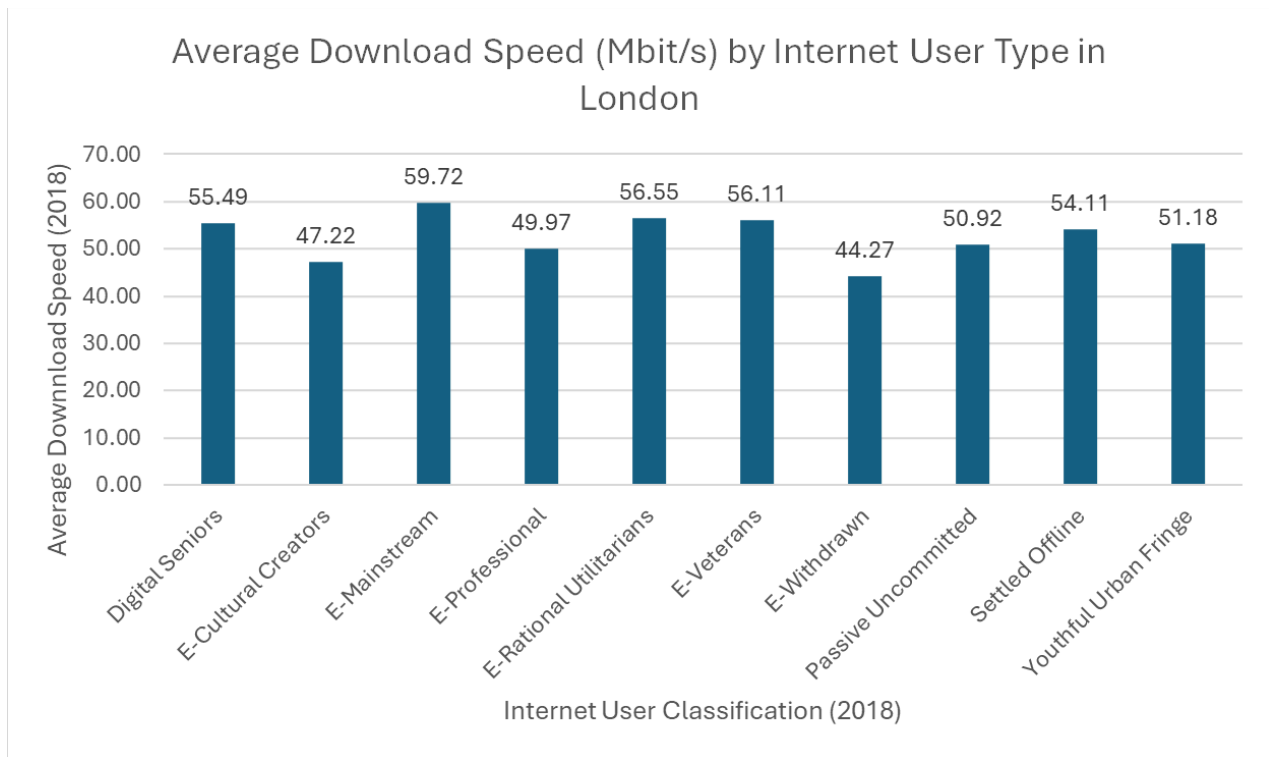
As seen in Figure 2, there is a concentration of e-Professionals in Central London, users defined as having the highest level of Internet engagement. These users are ethnically diverse and typically aged between 25 and 34, and use the Internet daily at their workplace and for communication, entertainment, gaming, and shopping. Moving out from Central London, there is a sizable portion of e-Veterans, representing affluent families, typically in low-density suburbs with populations of middle-aged and highly qualified professionals. These users are the second most common Internet users after e-Professionals, as they do not use the Internet for communication and entertainment.

Within greater London, there are pockets of Passive and Uncommitted Users, who have limited or no interaction with the Internet, and typically hold blue collar occupations. There are also regions with a high proportion of Youthful Urban Fringe, described as young, ethnic, living in informal households within materially deprived communities, and typically using the Internet via mobile devices due to a lack of access to desktop devices (Consumer Data Research Centre, 2018).

Comparing Figures 1 and 2, patterns emerge between broadband speed and Internet usage patterns. For example, the region of Southeast London with the slowest Internet speeds has a concentration of e-Rational Utilitarians: elderly, retired homeowners who view the Internet as a utility for basic tasks such as online grocery shopping. A surprising trend emerges in Central London, where pockets of low speeds

exist despite the area being prominently home to e-Professionals, Youthful Urban Fringe, and e-Veterans. This is possibly due to congestion in densely populated areas. In the following analysis, the average download speed by Internet user type will be examined.

Figure 3: Average Broadband Speed by Internet User Type



From Figure 3, E-Mainstream stands out with the highest average speed at nearly 60 Mbps. This group is described as the mode of Internet user characteristics and are often located at the periphery of urban areas. Meanwhile, E-Cultural Creators and Youthful Urban Fringe, despite being in central or high-demand areas, experience relatively lower speeds.

The Digital Seniors group, composed of retired, affluent white British users, has a high average, while the E-Withdrawn group experiences the lowest speed. This group is characterized by “less affluent white British individuals and areas of high ethnic diversity,” with the highest unemployment and social housing rates. Given that Digital Seniors and E-Withdrawn communities use the Internet with similar frequencies, it is possible that digital exclusion exists due to wealth disparities.

Broadband Speed and Deprivation Index

Seen in the previous analysis, there may be a socioeconomic explanation behind poor Internet access. I will compare the average download speed with the Index of Multiple Deprivation Decile, with a lower decile corresponding to greater deprivation. I reorganized the Broadband Speed data to group by LSOA to ensure comparability to the IMD Dataset, accounting for the slightly misshapen London boundaries.

Figure 4: Broadband Speed Averaged by LSOA

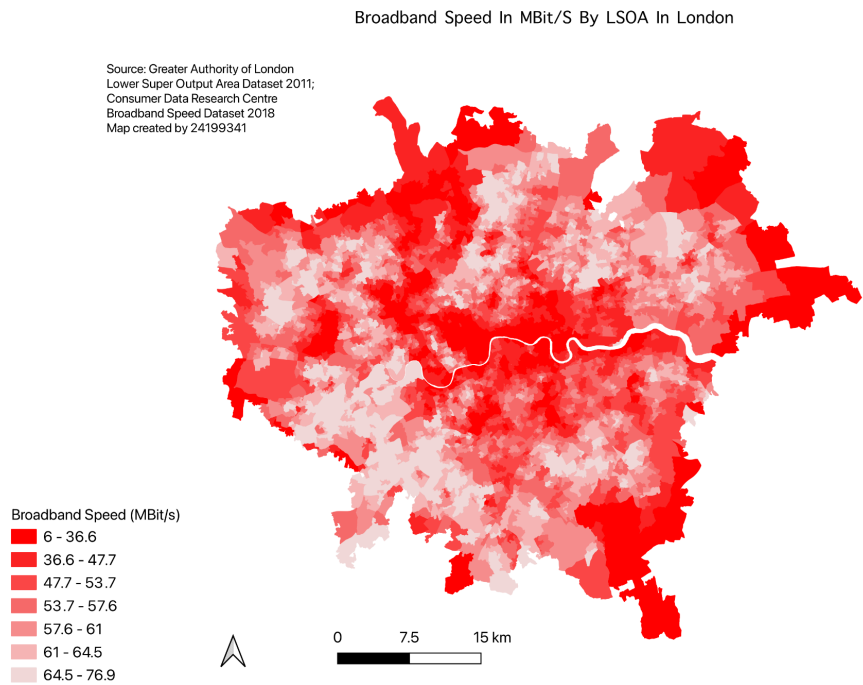
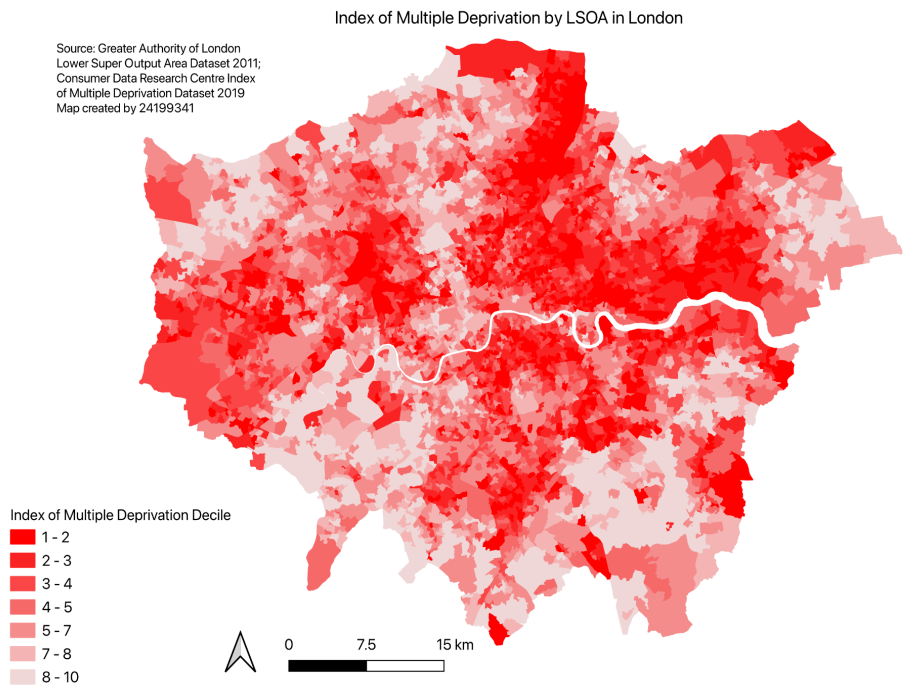
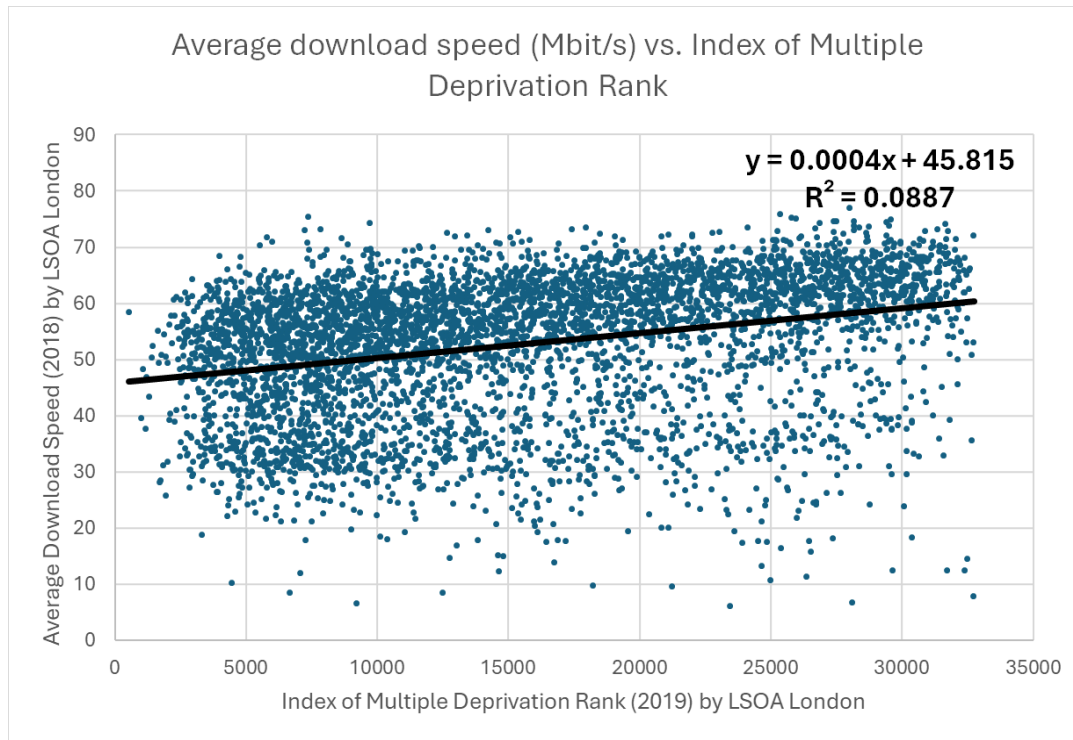


Figure 5: IMD Decile by LSOA



High concentrations of poor conditions, such as low broadband speeds and lower IMD deciles appear to be towards Central London. However, conclusions are not immediately apparent from these figures. I performed a regression analysis to determine if a correlation exists between broadband speed and IMD.

Figure 6: Scatterplot of Average Download Speed versus Index of Multiple Deprivation Rank



The weak correlation between IMD rank and broadband speed, with an R^2 value of only 0.0887 is surprising, but can be attributed to two main factors. First, the broadband speed data collected by Ofcom is based on sync speeds, which often overestimate real-world speeds due to network congestion and varying infrastructure quality. This methodology lacks the granularity to account for localized issues like congestion, maintenance, or interference, which can significantly affect actual user experience. Additionally, Ofcom's address-level data may not fully capture the specific characteristics of broadband service in small geographic areas, further distorting the speed metrics (Ofcom, 2018).

Secondly, aside from methodological issues, broadband investment may not follow a strict pattern. Areas with lower IMD ranks may still have good broadband speeds due to targeted investments, while wealthier urban areas could experience slower speeds due to congestion or outdated infrastructure.

The Root Cause: Public vs. Private Infrastructure

A digital divide enables systemic inequality, especially with the rise of online schooling and work models. In 2016, the United Nations declared Internet access as a basic human right under Article 19 of the Universal Declaration of Human Rights, emphasizing its role in education and employment access.

Private sector broadband deployment enables “digital redlining,” leaving older adults, people of color, and residents in most deprived regions underserved. Other countries are addressing similar issues: for example, the United States has invested in municipal broadband models. In 2010, Chattanooga, Tennessee launched a public utility Electric Power Board funded by \$111M in federal grants. This model offers speeds of 1Gbps, with affordable pricing programs for low-income households. The initiative attracted businesses, generated thousands of jobs, and improved healthcare and education access, making digital access a step to address greater issues (Sanders and Scanlon, 2021).

The UK is beginning to make strides in public broadband infrastructure. In October 2023, the UK government invested nearly £530 million to upgrade broadband networks in greater London, in collaboration with government agency Building Digital UK (BDUK). From their #GetOnlineWeek to promote social media activity, host public workshops at libraries on digital literacy, and lobby elected officials to invest in broadband infrastructure, BDUK has made significant strides (gov.uk).

The Internet must be reimagined as a human right rather than a luxury. Building on initiatives such as #GetOnlineWeek, the UK must provide targeted subsidies for low-income households and invest in rural connectivity.

Conclusion

This analysis examined the relationship between Internet speed, usage patterns, and deprivation, utilizing data from the Consumer Data Research Centre.

After comparing Internet speed and user classification, the speed appeared to correlate more strongly with the class of a population rather than with frequency of usage. For example, areas with elderly high-income residents who rarely use the Internet had higher broadband speeds than low-income areas where usage was similarly low.

However, mapping download speeds against IMD revealed no clear correlation. This discrepancy may be due to methodological mismatches in the data being analyzed. The broadband speed methodology may lack granularity and not capture the full picture of inequality.

In future studies, additional metrics could be explored, such as regional investment in broadband infrastructure, user-reported Internet performance, and access to digital literacy programs. These measures may provide a more nuanced understanding of the digital divide and its relationship with deprivation.

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